Keshav Shivkumar

keshavshivkumar8@gmail.com
https://linkedin.com/in/keshavshivkumar
https://github.com/keshavshivkumar
http://keshavshivkumar.me

EDUCATION

- **Master of Science (MS), Computer Science, 2024 | Rutgers University, New Brunswick, NJ, USA**
  CGPA: 3.83/4.00
  Relevant Coursework: Introduction to AI, Mathematical Foundations of Math Science, Machine Learning II, Computer Vision, Database Systems for Data Science
- **Bachelor of Technology (B.Tech), Computer Science Engineering, 2022 | PES University, Bangalore, India**
  Specialization: *Machine Intelligence and Data Science*
  CGPA: 8.11/10.00

PROFESSIONAL EXPERIENCE

Teaching Assistant, Rutgers University, Piscataway, New Jersey **Jan 2023 – Present**

- Served as the TA for CS-170, Computer Applications for Business, which involved teaching students basic concepts in HTML, CSS, JavaScript, pseudocode, MS Excel, and databases.
- Set up office hours to resolve student inquiries, graded assignments and exams, and conveyed complex technical concepts in an accessible and engaging manner.

Summer Intern, Bloom Energy, Bangalore, Karnataka **Sept 2021 – Mar 2022**

- Worked as a Summer Intern on the Development & DevOps team in a fast-paced Agile Scrum environment.
- Streamlined Bloom Energy's manufacturing services by implementing a Jenkins CI/CD pipeline triggered by Gitlab.
- Automated the Selenium Java test cases for the Bloom Energy manufacturing portal by developing a single-page web application (SPA) using Joomla CMS, which improved the efficiency and performance by 80%.
- Designed a proof-of-concept to trigger the Jenkins pipeline to run the selected test cases on the SPA by creating bash scripts that execute the Jenkins pipeline when a change is pushed to Gitlab.

TECHNICAL PROFICIENCIES

Programming Languages:	Python, Java, C, R
Web Development:	HTML, CSS, JavaScript, Django, ReactJS, Flask, Joomla, JSP
Other Relevant Skills:	SQL, Git, Linux, Data Analysis, PyTorch, OpenCV

PROJECTS

- **ViLT: Vision-and-Language Transformer** [Python (PyTorch)]: Contributed to the ICML 2021 conference paper "ViLT" where the model is finetuned on a novel dataset (GQA) offering valuable insights into the model's potential in different contexts.
- **Question Generation by Asking Questions** [Python (PyTorch)]: Implemented the EMNLP 2022 paper: *Learning to Generate Question by Asking Question: A Primal-Dual Approach with Uncommon Word Generation*, which which incorporates question generation and its dual problem, question answering into a unified primal-dual framework.
- **Shopfinity** [Java, SQL, JSP, JDBC, HTML, CSS, JavaScript]: Built a vehicle auctioning web application using Java and MySQL that allows users to create vehicle listings, place bids with an automatic bidding feature and ask questions through a forum.
- **Circle of Life** [Python]**:** Built probabilistic models of decision making for agents of different knowledge levels in a circular graph environment to catch prey while also avoiding a predator. Improved the models by using a Markov Decision Process and built a regression neural network model from scratch to ensure ~100% win rate in the different graph environments.
- **Cloud-Based Skin Cancer Detection Application** [Java, Python (TensorFlow, Keras), AWS]: Created a DenseNet CNN model trained using AWS SageMaker on the HAM10000 dataset stored on AWS S3 that classifies images of skin lesions and deployed it to AWS using API Gateway, with an Android application interface to allow users to send a skin lesion image.

PUBLICATIONS

- K. Shivkumar, G. Swaminathan, T. Puvvadi, A. Shekar and S. Shilpa, "Cloud-Based Model for Detecting Skin Cancer," 2022 12th International Conference on Cloud Computing, Data Science & Engineering (Confluence), 2022, pp. 106- 111, doi: 10.1109/Confluence52989.2022.9734157.